

$$\int_1^4 x + 1 \, dx$$

$$\Delta x = \frac{3}{n} \qquad f(a + i\Delta x) = 2 + i\Delta x \qquad \sum_{i=1}^n i = \frac{n * (n + 1)}{2}$$

$$\begin{aligned} A &= \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n (2 + i\Delta x) \Delta x \right] = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n 2\Delta x + i\Delta x^2 \right] = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n 2\Delta x + \sum_{i=1}^n i\Delta x^2 \right) = \\ &= \lim_{n \rightarrow \infty} \left(2\Delta x * n + \Delta x^2 \sum_{i=1}^n i \right) = \lim_{n \rightarrow \infty} \left(2\Delta x * n + \Delta x^2 * \frac{n * (n + 1)}{2} \right) = \\ &= \lim_{n \rightarrow \infty} \left(2 * \frac{3}{n} * n + \frac{9}{n^2} * \frac{n * (n + 1)}{2} \right) = 6 + \lim_{n \rightarrow \infty} \frac{9n + 9}{2n} = 6 + \frac{9}{2} = \frac{21}{2} \end{aligned}$$